3 Bulk Waves in Fluids

This chapter extends the introductory material of Chapter 2 to the simplest acoustic case of interest to us here, namely, the propagation of bulk waves in liquids and gases. Formally, this case is much simpler than that of solids; fluids in equilibrium are always isotropic, and only longitudinal (compressional) waves can propagate. Hence, there is no polarization to specify, and scalar wave theory can be applied. From another point of view, ultrasonic waves in liquids are sufficiently different from those in solids that a separate discussion is required. Finally, these results on liquids form a good basis for extending the theory to solids. A good discussion of waves in liquids is given in the study by Kinsler et al. [1] and Beyer and Letcher [2].

In terms of notation, V_i with a subscript i will be used for sound velocity, V_0 for bulk waves in liquids, V_L and V_S for longitudinal and shear waves in solids, respectively, V_P and V_G for phase and group velocities, respectively, and so on. When the symbol V stands alone, it normally represents the thermodynamic variable for volume V.

3.1 ONE-DIMENSIONAL THEORY OF FLUIDS

We consider bulk fluids that are homogeneous, isotropic, and compressible with equilibrium pressure p_0 and density ρ_0. As for the case with waves in strings in Chapter 2, we apply Newton's law to an element of volume, and we need an additional equation relating an increase in pressure to change in volume of the fluid, which will be provided by the definition of the compressibility.

Considering a simple volume element, a wave will be provided in the following way. If a pressure increase is applied at $t = 0$ to a plate at the origin, this will cause an increase in pressure and density in the layer of fluid next to it relative to the layer at the right. Hence, particles will flow to the right, leading to an increase in pressure and density, and the disturbance will then flow as a series of alternative compressions and rarefactions.

Considering the volume element between x and $x + \mathrm{d}x$, we have

$$\mathrm{d}F_x = \left[P(x) - \left\{P(x) + \frac{\partial P}{\partial x}\mathrm{d}x\right\}\right]A = -\frac{\partial P}{\partial x}\mathrm{d}xA \tag{3.1}$$

Applying Newton's law to the element of mass $\rho_0 \mathrm{d}xA$

$$\frac{\partial P}{\partial x} = -\rho_0 \frac{\partial^2 u}{\partial t^2} \tag{3.2}$$

Here, P and u are the instantaneous pressure and displacement, respectively.

For simplicity, we distinguish between the equilibrium pressure P_0 and the instantaneous pressure P to the excess, acoustic pressure p by

$$p = P - P_0$$

so

$$\frac{\partial p}{\partial x} = -\rho_0 \frac{\partial^2 u}{\partial t^2} \tag{3.3}$$

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To link the applied pressure to the compression of the liquid, we define the compressibility as follows:

$$\chi = -\frac{1}{V}\frac{\partial V}{\partial p} \tag{3.4}$$

and the compression of the liquid will be described by the dilatation S:

$$S \equiv \frac{V}{V} \tag{3.5}$$

During a compression of the volume $\mathrm{d}V = A\mathrm{d}x$ at pressure p on the left to $\mathrm{d}V = A\left(1 + \partial u/\partial x\right)\mathrm{d}x$ at pressure $p + \mathrm{d}p$ on the right

$$S = \frac{V}{V} = \frac{\partial u}{\partial x} \tag{3.6}$$

From the definition of the compressibility,

$$p = -\frac{S}{\chi} = -\frac{1}{\chi}\frac{\partial u}{\partial x} \tag{3.7}$$

Hence, the equation of motion can be rewritten as follows:

$$\frac{\partial^2 u}{\partial t^2} = V_0^2\frac{\partial^2 u}{\partial x^2} \tag{3.8}$$

where

$$V_0^2 \equiv \frac{1}{\rho_0\chi} \tag{3.9}$$

The compressibility can be rewritten as

$$\chi = -\frac{1}{V}\frac{\partial V}{\partial p} = \frac{1}{\rho_0}\frac{\partial \rho}{\partial p} \tag{3.10}$$

which gives a more general form

$$V_0^2 = \frac{\partial P}{\partial \rho} \tag{3.11}$$

Since pressure is only proportional to density in first order, this highlights the fact that $V_0 =$ constant only to first order. In other words, since the pressure–density relation is nonlinear in an exact theory, linear acoustics, corresponding to $V_0 =$ constant, does not exist as such but is only an approximation.

From Equations 3.2 to 3.11 the wave equation can be written in the following form.

$$\frac{\partial^2 u}{\partial t^2} = V_0^2\frac{\partial^2 u}{\partial x^2}$$

or

$$\frac{\partial^2 p}{\partial t^2} = V_0^2\frac{\partial^2 p}{\partial x^2} \tag{3.12}$$

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or

$$\frac{\partial^2 v}{\partial t^2} = V_0^2 \frac{\partial^2 v}{\partial x^2} \tag{3.13}$$

where

$v = \dfrac{\delta u}{\delta t}$ = particle velocity,
S = dilatation = $\partial u/\partial x$
$p = -\rho_0 V_0^2 S$

All of these three forms of the wave equation are equivalent by the above relations in the linear approximation. We will focus on the solutions for the displacement $u(x, t)$. These can be written as follows:

$$u = A \exp j(\omega t - kx) + B \exp j(\omega t + kx) = u_+ + u_- \tag{3.14}$$

where $A(u_+)$ is the amplitude (displacement) of the wave in the forward ($+x$) direction and $B(u_-)$ is the amplitude (displacement) of the wave in the backward ($-x$) direction.

Then, p, S, and v can also be written in the form

$$p = -\rho_0 V_0^2 \frac{\partial u}{\partial x} = j\rho_0 \omega V_0 (u_+ - u_-) \tag{3.15}$$

$$S = \frac{\partial u}{\partial x} = jk(-u_+ + u_-) \tag{3.16}$$

$$v = \frac{\partial u}{\partial t} = j\omega(u_+ + u_-) \tag{3.17}$$

One immediate consequence of these equations is that they provide the phase relations between pressure, displacement, dilatation, and velocity. These can best be displayed on a complex phasor diagram, as shown in Figure 3.1. From a practical viewpoint, the relation for pressure and velocity is most important. For the forward wave, pressure and velocity lead the displacement by $\pi/2$; for the backward wave, velocity leads by $\pi/2$ and pressure lags by $\pi/2$. The change in phase relationship with propagation direction comes about because pressure and dilatation are scalar quantities, whereas displacement and velocity are vectorial.

3.1.1 Sound Velocity

As seen by the form of the solutions, the sound velocity $V_0 = \sqrt{\partial P/\partial \rho} = \omega/k$ is the phase velocity of the wave. For bulk waves in infinite media, it is a constant for a given medium but is dependent on all of the thermodynamic parameters such as compressibility, density, external pressure, and temperature. Within the present context, it is independent of frequency (infinite media) and amplitude (linear regime), but in general, this is, of course, not the case. In fact, the analysis of the velocity is quite different for gases and liquids, and therefore, these two cases will be treated separately.

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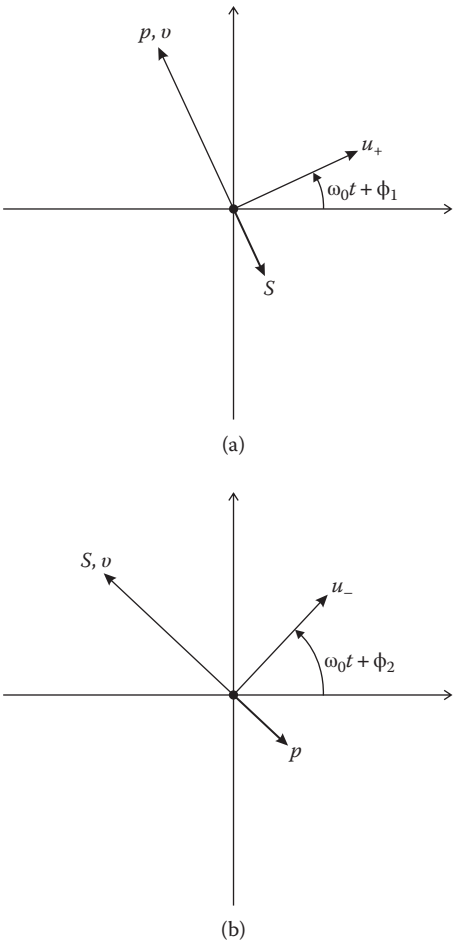

FIGURE 3.1 Phasor representation for an acoustic wave in a fluid. (a) Forward wave. (b) Backward wave.

3.1.1.1 Gases

The approximation of an ideal gas will be made as follows: $PV = n_0RT$ or $P = (RT/M)\rho$, where n_0 = number of moles. Since sound propagation in a gas is known to be essentially an adiabatic process, the relation $PV\gamma$ = constant is also applicable. This can be written in the form

$$\frac{P}{\rho^{\gamma}} = \text{constant} \tag{3.18}$$

so that

$$\frac{\partial P}{\partial \rho} = \frac{\gamma P}{\rho} \tag{3.19}$$

and for equilibrium conditions P_0, ρ_0

$$V_0 = \sqrt{\frac{\gamma P_0}{\rho_0}} = \sqrt{\frac{\gamma RT}{M}} \tag{3.20}$$

For air (diatomic) at room temperature (20°C), $\gamma = 1.4$ and $P_0 = 1.01 \times 10^5$ Pa, giving $V_0 \sim 343$ m/s in good agreement with experiment.

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In the present treatment of fluids, the first implicit assumption of local thermodynamic equilibrium is that only under this condition can local values of P, T, ρ, and so on be assigned. In the case of a gas, the length scale for thermodynamic equilibrium is the mean free path l of the gas particles, that is, the mean distance between collisions of the molecules. It is standard that

$$l = v_0 \tau \tag{3.21}$$

where

τ = mean time between collisions
v_0 = thermodynamic particle velocity of the molecules
l can be inferred from transport measurements on the gas, and v_0 is well known from the kinetic theory of gases. In order of magnitude, $v_0 = \sqrt{3RT/M} \propto 300$ m/s at 20°C.

The second implicit assumption is that to obtain wave propagation conditions, the thermodynamic parameters must be well defined over distances much shorter than the wavelength. Otherwise, the propagating quantities such as pressure and density would simply not be defined with respect to the wave. This then gives the condition $\lambda \gg l$, which must be satisfied for a wave description to apply. This implies an upper frequency limit for wave propagation in a gas, for example, at STP for air $l \propto 10^{-5}$ cm, leading to a critical frequency $f \propto 1$ GHz.

It should be noted that the same conditions apply for liquids and solids, but the critical frequencies are much higher and do not have any practical consequence for ultrasonic waves.

3.1.1.2 Liquids

It is relatively easy to find simple models for the limiting cases of sound propagation in gases and solids. Liquids, however, constitute an intermediate case, and it is more difficult to find a simple model connecting the sound velocity V_0 to the molecular constants. The few available models will be outlined briefly.

A semiempirical approach, similar to that for gases, gives

$$V_0 = \sqrt{\frac{\gamma K_T}{\rho_0}} \tag{3.22}$$

where K_T is the isothermal bulk modulus. Another semiempirical approach is Rao's rule, which is the form

$$V_0^{1/3} V = R_a \tag{3.23}$$

where V is the molar volume and R_a is a constant for a given liquid. It was pointed out by Rao that R_a undergoes regular increments among the members of a homologous series of liquids so that

$$R_a = AM + B \tag{3.24}$$

where M is the molecular weight.

One of the few relations between V_0 and the liquid structure was provided by the early study of Schaaffs [3]. He assumed that although a realistic equation of state for the liquid was too complicated, some properties of organic liquids such as the sound velocity could be deduced from the van der Waals equation:

$$P + \frac{a}{V^2}\ (V - b) = RT \tag{3.25}$$

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where

R = the universal gas constant
a = constant
b = excluded volume

For organic liquids Schaaffs obtained

$$V_0 = \sqrt{\gamma RT \quad \frac{M}{3(M-\rho b)^2} - \frac{2}{M-\rho b}} \tag{3.26}$$

Actual comparisons were made by solving for b

$$b = \frac{M}{\rho} \quad 1 - \frac{RT}{MV_0^2} \quad 1 + \frac{MV_0^2}{3RT} \quad ^{\frac{1}{2}} - 1 \tag{3.27}$$

Excellent agreement was obtained by comparing $b = 4V_{\text{molecule}}$ with molecular volumes determined by other means. Further discussion of other semiempirical approaches is given by Beyer and Letcher [2], including that for the sound velocity in liquid mixtures. Values for representative liquids are given in Table 3.1.

3.1.2 Acoustic Impedance

Using the electromechanical analogy developed in Chapter 2, we define the specific acoustic impedance Z of an acoustic wave as follows:

$$Z = \frac{p}{v} \tag{3.28}$$

Z carries a sign as v can be in either the positive or negative direction. The absolute value of Z for plane waves, useful to characterize the bulk (infinite) medium, is called the characteristic impedance of the liquid, $Z_0 = \rho_0 V_0$. A third variant, the normal acoustic impedance, will be introduced in Chapter 7 for reflection and transmission analysis.

TABLE 3.1
Acoustic Properties of Representative Liquids

Liquid	V_L (km·s^{-1})	ρ (10^3 kg·m^{-3})	Z_0 (MRayls)
Acetone	1.17	0.79	1.07
Liquid argon (87K)	0.84	1.43	1.20
Methanol	1.1	0.79	0.87
Gallium (30K)	2.87	6.10	17.5
Glycerin	1.92	1.26	2.5
Liquid He^4 (2K)	0.228	0.145	0.033
Mercury	1.45	13.53	19.6
Liquid nitrogen (77K)	0.86	0.85	0.68
Silicone oil	1.35	1.1	1.5
Seawater	1.53	1.02	1.57
Water (20°C)	1.48	1.00	1.483

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Using the previous notation, we can determine the acoustic impedance for forward and backward propagations:

$$Z_+ = \frac{p_+}{v_+} = \frac{j\rho_0\omega V_0 u_+}{j\omega u_+} = \rho_0 V_0 \tag{3.29}$$

$$Z_- = \frac{p_-}{v_-} = \frac{-j\rho_0\omega V_0 u_-}{j\omega u_-} = -\rho_0 V_0 \tag{3.30}$$

Acoustic impedance is a highly useful concept in ultrasonics. From Chapter 2, it is the direct analogy of impedance in electrical circuits. In the case of electrical circuits, it is well known that there is maximum power transfer between two circuits when the impedances are matched. In the ultrasonic case, this corresponds to maximum transmission of an ultrasonic wave from one medium to another when the characteristic impedances are equal. Characteristic acoustic impedances for some liquids are shown in Figure 3.2 in a representation that is useful for choosing liquids with prescribed density and sound velocity.

3.1.3 Energy Density

The energy density is the total energy per unit volume, the sum of the kinetic and potential energy. By definition, the kinetic energy density is

$$u_k = \rho_0 \dot{u}^2 \tag{3.31}$$

For potential energy, we consider a volume element V changed to V' by the passage of the acoustic wave.

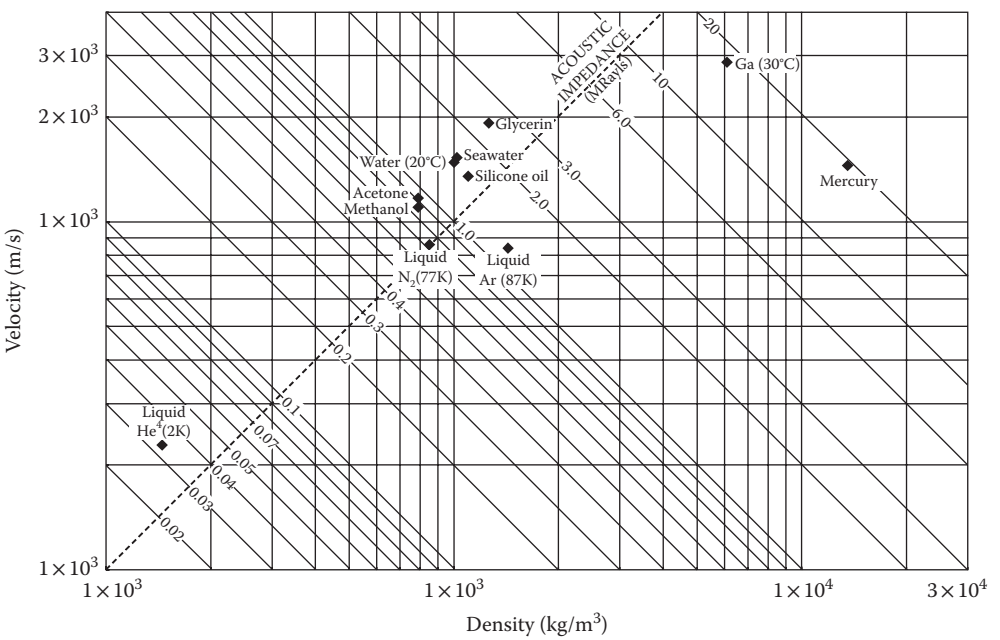

FIGURE 3.2 Density–sound velocity/characteristic acoustic impedance relation on a log–log scale for various liquids. (Based on a graph by Eggleton, R. C., described in Jipson, V. B., "Acoustical Microscopy at Optical Wavelengths," PhD thesis, E. L. Ginzton Laboratory, Stanford University, Stanford, CA, 1979.)

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Since $S = \dfrac{\partial u}{\partial x}$ from Equation 3.6

$$\begin{aligned} V &= V \quad 1 + \frac{\partial u}{\partial x} \\ &= V \quad 1 - \frac{p}{\rho_0 V_0^2} \end{aligned} \tag{3.32}$$

and the change in potential energy is

$$U_P = -\int p \mathrm{d}V \tag{3.33}$$

From Equation 3.32

$$\mathrm{d}V = -\frac{V \mathrm{d}p}{\rho_0 V_0^2} \tag{3.34}$$

Hence,

$$U_P = \frac{V}{\rho_0 V_0^2} \int_0^p p \mathrm{d}p = \frac{1}{2} \frac{p^2 V}{\rho_0 V_0^2} \tag{3.35}$$

Finally,

$$U_{\mathrm{tot}} = \quad U_{\mathrm{K}} + \quad U_{\mathrm{P}} = \frac{1}{2} \rho_0 \quad \dot{u}^2 + \frac{p^2}{\rho_0^2 V_0^2} \quad V \tag{3.36}$$

so that the acoustic energy density

$$u_{\mathrm{a}} = \frac{U_{\mathrm{tot}}}{V} = \frac{1}{2} \rho_0 \quad \dot{u}^2 + \frac{p^2}{\rho_0^2 V_0^2} \tag{3.37}$$

3.1.4 Acoustic Intensity

The acoustic intensity I is the average flux of acoustic energy per unit area per unit time. For a plane wave, it is clear that for a tube element of area A and length $V_0 \mathrm{d}t$, all of the acoustic energy $\mathrm{d}U_{\mathrm{a}}$ inside the cylindrical element will traverse the end face and leave the cylinder in time $\mathrm{d}t$.

Hence,

$$\mathrm{d}U_{\mathrm{a}} = u_{\mathrm{a}} A V_0 \, \mathrm{d}t$$

so that

$$I \equiv \frac{\overline{\mathrm{d}U_{\mathrm{a}}}}{A \mathrm{d}t} = \bar{u}_{\mathrm{a}} V_0 \tag{3.38}$$

3.2 THREE-DIMENSIONAL MODEL

The previous results can be generalized immediately to three dimensions. Displacement u and velocity v now become explicitly vectors $\vec{u}$ and $\vec{v}$, whereas the acoustic pressure p remains a scalar. Hence, the 3D description of the acoustic properties of fluids is usually carried out in terms of the acoustic pressure; not only is this the simplest choice, but pressure is also the variable that is usually measured in the laboratory.

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For a surface element $\overrightarrow{\mathrm{d}A}$ with displacement $\vec{u}$, the associated volume is $\mathrm{d}V = \vec{u} \bullet \overrightarrow{\mathrm{d}A}$. By Gauss' theorem

$$V = \oint_S \vec{u} \bullet \overrightarrow{\mathrm{d}A} = \int_V (\vec{\ } \bullet \vec{u})\,\mathrm{d}V \equiv \int_V S(\vec{r})\,\mathrm{d}V \tag{3.39}$$

where $S(\vec{r})$ is the dilatation.

Hence,

$$S(\vec{r}) = \vec{\ } \bullet \vec{u} \equiv \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y} + \frac{\partial u_z}{\partial z} \tag{3.40}$$

Newton's law in three dimensions is

$$\rho_0 \frac{\partial^2 \vec{u}}{\partial t^2} = -\vec{\ } p \tag{3.41}$$

where $-\vec{\ } p$ is the net force on the element.

We want to change to a simple set of variables so that $\vec{u}$ on the left-hand side should be expressed in terms of the pressure. This can be done by using $S(\vec{r}) = \vec{\ } \bullet \vec{u}$ and then using Equation 3.7, the relation between the dilatation and the pressure. By applying those steps to Equation 3.41, we obtain

$$\rho_0 \frac{\partial^2 S}{\partial t^2} = - \ (p) \tag{3.42}$$

where

$$= \vec{\ } \bullet \vec{\ } = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \equiv \text{Laplacian}$$

and finally the wave equation

$$(p) = \frac{1}{V_0^2} \frac{\partial^2 p}{\partial t^2} \tag{3.43}$$

where

$$V_0^2 = \frac{1}{\rho_0 \chi} \tag{3.44}$$

In analogy with Equation 3.43, the 3D wave equations for $\vec{u}$ and $\vec{v}$ are

$$^2\vec{u} = \frac{1}{V_0^2} \frac{\partial^2 \vec{u}}{\partial t^2} \tag{3.45}$$

$$^2\vec{v} = \frac{1}{V_0^2} \frac{\partial^2 \vec{v}}{\partial t^2} \tag{3.46}$$

and the solutions for $\vec{u}$ are

$$\vec{u} = \vec{u}_0 \exp j(\omega t - \vec{k} \bullet \vec{r}) \tag{3.47}$$

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where $\vec{k}$ is the propagation vector whose direction gives the direction of propagation and whose magnitude is

$$\left|\vec{k}\right| = \frac{2\pi}{\lambda} \tag{3.48}$$

3.2.1 Acoustic Poynting Vector

In the presence of applied volume forces $\vec{f}$ per unit volume, Equation 3.42 becomes

$$\rho_0 \frac{\partial \vec{v}}{\partial t} = -\vec{\nabla}(p) + \vec{f} \tag{3.49}$$

If this force represents the force by the adjoining fluid on an element dV, then the work done per unit volume in time dt is

$$\begin{aligned} \mathrm{d}w &= \vec{f} \bullet \overrightarrow{\mathrm{d}u} = \vec{f} \bullet \vec{v}\mathrm{d}t \\ &= \rho_0 \vec{v} \bullet \overrightarrow{\mathrm{d}v} + \vec{\nabla} p \bullet \overrightarrow{\mathrm{d}u} \text{ by Equation 3.49} \\ &= \mathrm{d}\left(\frac{1}{2}\rho_0 \vec{v}^2\right) - p\mathrm{d}S + \vec{\nabla} \bullet (p\overrightarrow{\mathrm{d}u}) \end{aligned} \tag{3.50}$$

Referring to the 1D model, we immediately identify the first two terms as the variation of the kinetic and potential energy per unit volume, respectively.

Hence,

$$u_{\mathrm{K}} = \frac{1}{2}\rho_0 v^2 \tag{3.51}$$

and

$$u_{\mathrm{P}} = -\int_0^S p\mathrm{d}S = \int_0^S \frac{S\mathrm{d}S}{\chi} = \frac{1}{2}\frac{S^2}{\chi} \quad \text{by Equation 3.7}$$

We define the acoustic Poynting vector

$$\vec{\mathcal{P}} \equiv p\vec{v} \tag{3.52}$$

and take the time derivative of Equation 3.50

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(u_{\mathrm{K}} + u_{\mathrm{P}}) + \vec{\nabla} \bullet \vec{\mathcal{P}}$$

For a finite system, we integrate over the volume:

$$\frac{\mathrm{d}w}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}(U_{\mathrm{K}} + U_{\mathrm{P}}) + \oint_S \vec{\mathcal{P}} \bullet \overrightarrow{\mathrm{d}A} \tag{3.53}$$

where $\vec{\mathcal{P}}$ is the instantaneous acoustic power per unit area radiated from the system through the surface S. Equation 3.53 represents the law of conservation of energy at a given time.

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The average value of $\vec{\mathcal{P}} \equiv \vec{I}$ then corresponds to the average flux density carried by the acoustic wave. For a system with no absorption $\vec{I}$ = constant and by Equation 3.53, the net acoustic power radiated from a closed element in the steady state is zero.

3.2.2 Attenuation

Up to now, we have assumed perfectly lossless reversible behavior of the fluid. In practice, there are losses or absorption of acoustic energy by the medium. These losses are normally attributable to viscosity and thermal conductivity, leading to the so-called classical attenuation. In addition, there are molecular processes in which acoustic energy is transformed into internal molecular energy. The finite time for these processes leads to relaxation and loss effects.

In fact, all of the loss effects in fluids can be described by a phase lag between acoustic pressure and medium response (density or volume change). A classical example from thermodynamics is the *P–V* diagram, which can be used to display the work done on a medium due to a pressure change.

The situation is shown in Figure 3.3 on the usual *P–V* diagram for compression and expansion of a gas. Let us suppose that changes in *P* and *V* are due to an acoustic wave. The work done or supplied by the system is given by

$$W = -\int P \mathrm{d}V$$

for the appropriate process. It is well known that the area enclosed by the curve for a cycle is the net work done on the system. In the lossless case, the system evolves along the same path *I* during expansion from *A* to *B* ($\underset{A}{\overset{B}{\equiv}}$) and compression from *B* back to *A* ($\underset{B}{\overset{A}{\equiv}}$). These two amounts of work are of opposite sign, and therefore, the net amount of work absorbed by the system from the acoustic wave is zero. On the other hand, if the system does not respond immediately, then intuitively change in volume will tend to lag that for the reversible case for both expansion (II) and compression (I), leading to a net amount of work per cycle by the acoustic wave on the medium, leading to absorption of energy.

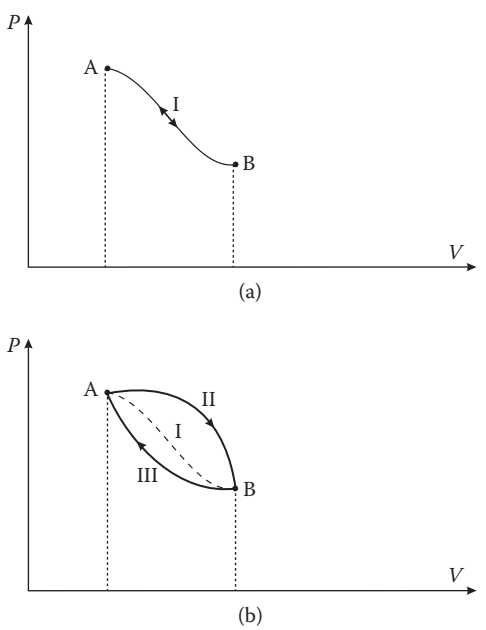

FIGURE 3.3 (a) Reversible transformation from *A* to *B* and from *B* to *A* in a lossless medium. (b) Transformation from *A* to *B* and from *B* to *A* in a lossy medium.

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3.2.2.1 Decibel Scale of Attenuation

If we consider the displacement u of the wave as

$$u = u_0 \exp j(\omega t - kx)$$

for the wave without dissipation, then $I \propto u^2$ for plane waves. If now we add dissipation, the only effect is that the wave vector $\vec{k}$ becomes complex, that is, $k \to \beta - j\alpha$, where α is seen to be the attenuation coefficient for the amplitude of the wave, as now

$$u = u_0 \exp j(\omega t - \beta x) \exp(-\alpha x) \tag{3.54}$$

In plane wave conditions, which are standard for attenuation measurement, $I \propto u^2$, so that the acoustic intensity decays as $\exp(-2\alpha x)$. The factor of two comes from the difference in attenuation between the amplitude and the intensity due to the quadratic term. In practice, care must be taken as to what is being measured (and calculated) to avoid confusion on this point.

In practice, the attenuation factor for the amplitude is measured by determining the amplitude ratio r_{12} of the wave at two different positions x_1 and x_2.

Hence,

$$r_{12} = \exp \alpha(x_2 - x_1)$$

The attenuation in nepers $\equiv \ln(r_{12}) = \alpha(x_2 - x_1)$, so that α is measured in Np/m.

It is more common to use the decibel (dB) scale to compare acoustic intensity level; the attenuation in dB is defined as

$$\begin{aligned} \text{attenuation (dB)} &= 10 \log_{10}(r_{12})^2 \\ &= 20(\log_{10} \mathrm{e})\alpha(x_2 - x_1)\text{dB} \end{aligned} \tag{3.55}$$

where α is in dB/m.

Hence, the relation between the two units is

$$\alpha \text{ (dB/m)} = 20(\log_{10} \mathrm{e})\alpha \text{ (Np/m)} = 8.686\alpha \text{ (Np/m)} \tag{3.56}$$

3.2.2.2 Relaxation Time Formulation for Viscosity

Stokes' classic treatment includes a time-dependent term in the pressure–condensation relation [1]

$$p = \rho_0 V_0^2 S + \eta \frac{\partial S}{\partial t} \tag{3.57}$$

where η is a viscosity coefficient and $s = -S$ is the relative density change or condensation. For an applied pressure $p_a = p_{a0} \exp(j\omega t)$, if we assume a response for the condensation $s = s_0 \exp(j\omega t)$, direct substitution yields

$$S_0 = \frac{p_0}{\rho_0 V_0^2 + j\omega\eta} \tag{3.58}$$

Clearly, the density change lags the applied pressure by a phase angle ϕ, where

$$\tan \phi = \frac{\omega\eta}{\rho_0 V_0^2} \tag{3.59}$$

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If a step function pressure change Δp_{a0} is applied at $t = 0$, the solution is

$$S = \frac{p_0}{\rho_0 V_0^2} \quad 1 - \exp \quad -\frac{\rho_0 V_0^2 t}{\eta} \tag{3.60}$$

and if a step function pressure is suddenly removed,

$$S = \frac{p_0}{\rho_0 V_0^2} \exp \quad -\frac{\rho_0 V_0^2 t}{\eta} \tag{3.61}$$

Recalling the electromechanical analogy, it is readily seen that these solutions are identical to those for the current in an L–R circuit when a potential difference is suddenly applied or removed. That process is described by a relaxation time $\tau = L/R$. By analogy, we define a viscous relaxation time

$$\tau = \frac{\eta}{\rho_0 V_0^2} \tag{3.62}$$

3.2.2.3 Attenuation Due to Viscosity

The effects of attenuation are normally incorporated by using a complex wave number.

$$k \equiv \beta - j\alpha \tag{3.63}$$

Then

$$u = u_0 \exp j(\omega t - (\beta - j\alpha)x) = u_0 \, e^{-\alpha x} \exp j(\omega t - \beta x) \tag{3.64}$$

using the Stokes term for the pressure, the wave equation is

$$\frac{\partial^2 u}{\partial t^2} = V_0^2 \frac{\partial^2 u}{\partial x^2} + \frac{\eta}{\rho_0} \frac{\partial^2 u}{\partial x \partial \tau} \tag{3.65}$$

substituting for u and separating the real and imaginary parts

$$\alpha^2 = \frac{\omega^2}{2V_0^2} \quad \frac{1}{\sqrt{1 + \omega^2\tau^2}} - \frac{1}{1 + \omega^2\tau^2} \tag{3.66}$$

$$\beta^2 = \frac{\omega^2}{2V_0^2} \quad \frac{1}{\sqrt{1 + \omega^2\tau^2}} + \frac{1}{1 + \omega^2\tau^2} \tag{3.67}$$

and

$$V_P^2 \equiv \frac{\omega^2}{\beta^2} = \frac{2V_0^2(1 + \omega^2\tau^2)}{1 + \sqrt{1 + \omega^2\tau^2}} \tag{3.68}$$

For most fluids at ultrasonic frequencies at room temperature, $\omega\tau \ll 1$; hence,

$$\alpha \sim \frac{\omega^2\tau}{2V_0} = \frac{\omega^2\eta}{2\rho_0 V_0^3} \tag{3.69}$$

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and the modified phase velocity

$$V_P = V_0 \quad 1 + \frac{3}{8}\omega^2\tau^2 \tag{3.70}$$

The important result here is that in this limit $\alpha \sim \omega^2$. This means that α rises rapidly with frequency; this will have important implications for acoustic devices and nondestructive evaluation (NDE). The change in velocity is small and is neglected in most cases in practice.

3.2.2.4 Attenuation Due to Thermal Conduction

In simple descriptions of sound propagation, perfect adiabaticity is usually assumed. This is only strictly true if the thermal conductivity $k \equiv 0$. In fact, there is always a finite κ, and therefore, heat will be transported from the hot regions (compressions) to the cooler regions (rarefactions) created by the sound wave. As for viscous effects, the temperature change will lag the applied pressure, leading to additional attenuation, described by a relaxation time

$$\tau = \frac{\kappa}{\rho_0 V_0 C_p} \tag{3.71}$$

The corresponding attenuation in the limit $\omega\tau \ll 1$ when added to the viscous term that Equation 3.69 gives for the so-called classical attenuation coefficient of liquids.

$$\alpha = \frac{\omega^2}{2\rho_0 V_0^3} \quad \frac{4\eta}{3} + \frac{\kappa(\gamma - 1)}{C_p} \tag{3.72}$$

It is interesting to compare this classical attenuation to that actually observed experimentally in liquids, which is done in Table 3.2 for liquids and gases. Excellent quantitative agreement is

TABLE 3.2
Acoustic Absorption in Fluids

All Data for T = 20°C and $\mathcal{P}_0$ = 1 atm	α/f^2 ($Np \cdot s^2 \cdot m^{-1}$)			
	Shear Viscosity	Thermal Conductivity	Classical	Observed
Gases		**Multiply all values by 10^{-11}**		
Argon	1.08	0.77	1.85	1.87
Helium	0.31	0.22	0.53	0.54
Oxygen	1.14	0.47	1.61	1.92
Nitrogen	0.96	0.39	1.35	1.64
Air (dry)	0.99	0.38	1.37	α/f peaks at 40 Hz
Carbon dioxide	1.09	0.31	1.40	α/f peaks at 30 kHz
Liquids		**Multiply all values by 10^{-15}**		
Glycerin	3000.0	—	3000.0	3000.0
Mercury	—	6.0	6.0	5.0
Acetone	6.5	0.5	7.0	30.0
Water	8.1	—	8.1	25.0
Seawater	8.1	—	8.1	α/f peaks at 1.2 kHz and 136 kHz

Source: Data from Kinsler, L. E. et al., *Fundamentals of Acoustics*, John Wiley & Sons, New York, 2000.

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obtained for inert gases (He and Ar) and for cases in which the viscous term dominates (glycerin). Otherwise, the experimental value exceeds the classical one sometimes significantly. This is due to molecular relaxation phenomena.

3.2.2.5 Molecular Relaxation

This is a subject of physical chemistry in itself, which could easily fill a large book. However, since the subject is now well understood and is not of current research interest, only a brief overview will be given. A detailed discussion has been given by Herzfeld and Litovitz [4] and Beyer and Letcher [2].

To see how molecular structure can contribute to relaxation effects, let us look briefly at the simple physics of relaxation, which will also give insight into the viscosity and thermal conduction contributions. Consider a physical system at constant temperature that is excited to a higher energy state by energy absorbed from an incident ultrasonic wave. The system will attempt to return to equilibrium by giving up this energy to the surrounding regions at a rate determined by a temperature-dependent relaxation time. Let us now slowly increase the ultrasonic frequency from zero.

In the regime $\omega\tau \ll 1$, the variations of the applied field are so slow that the process is approximately reversible, the system follows in phase with the applied field, there is little excess absorbed energy, and the attenuation is tiny. This is less true as the frequency is increased, leading to an increase in attenuation.

In the opposite limit with $\omega\tau \gg 1$ at sufficiently high frequencies, the ultrasonic field varies so fast that the system cannot follow it. Hence, there is almost no absorbed energy and the attenuation is again very small. As the frequency is reduced, the system progressively starts to follow the field and absorb energy; hence, the attenuation starts to increase.

Clearly, there is an optimal situation at which the system absorbs a large amount of energy from the field and dumps it efficiently and irreversibly as heat to the surroundings, thus giving a high value of attenuation. This optimal situation occurs at $\omega\tau \sim 1$, at which there is a well-defined peak in the attenuation. Experimentally, we can also observe the peak at constant frequency by varying the temperature (or other parameter), which makes τ sweep the critical region around the peak. Another point is that experimentally the relaxation peak is very sharp and may be confused with a resonance; care must be taken in interpretation as the physics in the two cases is quite different.

The case of gases is the simplest to analyze. The monatomic and inert gases have only translational degrees of freedom. τ is hence very short, and there is no excess attenuation above the "classical" value. Polyatomic gases have rotational and vibrational levels that require a finite time to take up the excess energy, particularly the latter. This leads to a specific heat of the form

$$C_V = C_e + C_m \quad 1 - e^{-\frac{t}{\tau}} \tag{3.73}$$

which will obviously lead to a relaxation type attenuation. A classic example is CO_2, which at STP exhibits a relaxation peak centered at about 20 kHz; the maximum attenuation is of the order of 1200 times that of the classical value. Another interesting example is air. Dry air exhibits slightly greater attenuation than the classic value, but for humid air below 100 kHz, there is an order of magnitude increase. This is due to the reduction in τ for the vibrational mode of the oxygen molecule due to the catalytic effect of the water vapor molecules.

Similar relaxation effects have been found in liquids, and this explains the excess attenuation in nonpolar liquids such as acetone. The excess attenuation observed in water has been shown to be due to structural relaxation as explained by Hall [5]. Water is known to be a two-state liquid, part of the water in a free state, the rest in a bound state in which the water molecules have a more closely

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packed structure. The ultrasonic wave causes transitions between the two states, and the associated time delays lead to a relaxation-type phenomenon. An additional excess attenuation occurs in seawater where dissolved salts lead to a type of chemical relaxation.

SUMMARY

Wave equation is a second-order differential equation that allows determination of the displacement $u(x, t)$ for given initial and boundary conditions.

Sound velocity in a gas is given by the general formula $V_0^2 = dP/d\rho$. For a perfect gas, $V_0 = \sqrt{\gamma RT/M}$.

Sound velocity in a liquid to a first approximation is given by $\sqrt{\gamma K_T/\rho_0}$, where K_T is the isothermal bulk modulus.

Specific acoustic impedance $Z = p/v$ carries a sign; it is positive in the forward direction and negative in the backward direction. The absolute value for plane waves is the characteristic acoustic impedance $Z_0 = \rho_0 V_0$ and is a constant for the medium.

Acoustic intensity is the average flux of acoustic energy per unit area per unit time. For a plane wave, it is given by $\overline{u_a} V_0$.

Acoustic Poynting vector is defined as $p\vec{v}$ and is the flux density of acoustic energy in a given direction.

Decibel is a log scale used to compare acoustic intensities. Acoustic attenuation in a medium is expressed in dB/m or Np/m.

Classical attenuation for liquids and gases is due to viscosity and thermal conduction.

Molecular relaxation occurs in polyatomic gases and liquids. Because of the phase lag in transferring the ultrasonic energy to the different energy levels, relaxation gives rise to an extra attenuation, which is described by the parameter $\omega\tau$. The limit $\omega\tau \ll 1$ applies to most media at ultrasonic frequencies.

QUESTIONS

1. Draw waveforms as a function of x for $u(x)$, $v(x)$, and $P(x)$ for a traveling harmonic wave. Comment on the phase relationships in the forward direction of the form in Equation 3.14.
2. Use the results of Question 1 and Equation 3.52 to calculate $I(x)$ for this wave. Calculate and sketch the graph of both instantaneous and average values of $I(x)$.
3. Using data from Table 3.1, calculate for glycerin:
   i. Viscous relaxation time τ_s
   ii. Low-frequency attenuation α_s
   iii. Compare this result with that of Table 3.1
4. Use an approximation for air as a perfect gas of molecular weight 29. At STP (0°C and 1 atmosphere of pressure), calculate the following:
   i. Mass density
   ii. Average molecular velocity
   iii. Mean free time between collisions
   iv. Mean free path between collisions
5. Justify Equation 3.7.
6. Show that the condensation s, the density change per unit density $= -S$, where S is the dilatation.
7. Find the specific acoustic impedance for a standing wave $p = p_0 \sin kx \cos\omega t$.
8. For two waves of different frequency traveling in the $+x$ direction, show that the specific acoustic impedance is $\rho_0 V_0$.

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